

Kha. Mo. Syeed Asif

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Aftabnagar Dhaka, Bangladesh

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate with strong analytical, research, and problem-solving skills. Proficient in Python, SQL, Microsoft Excel, Git, and Linux, with experience in data analysis and web technologies. A quick learner with excellent communication skills and a keen interest in Search Engine Optimization (SEO) and digital marketing. Eager to apply technical knowledge, learn industry best practices, and contribute effectively in a collaborative environment.

TECHNICAL SKILLS

Programming	Python, SQL, JavaScript, C/C++
SEO & Digital Marketing	Keyword Research, Content Optimization, Link Building (Basic), Website Analysis
Data Analysis	Microsoft Excel, Data Cleaning, Data Analysis, Exploratory Data Analysis (EDA), KPI Reporting, Data Visualization
Web Technologies	HTML, CSS, JavaScript, Responsive Web Design, Basic Website Structure
AI & Automation	Prompt Engineering, AI-Assisted Content Creation, ChatGPT, Basic Machine Learning
Libraries & Frameworks	Pandas, NumPy, Scikit-learn
Databases	MySQL, Oracle SQL
Tools	Git, GitHub, Linux, Microsoft Office, Power BI, Google Docs, Google Sheets
Soft Skills	Analytical Thinking, Research, Problem Solving, Communication, Teamwork, Time Management, Quick Learner

EDUCATION

B.Sc. in Computer Science & Engineering (Major: Data Science) East West University, Dhaka	2022–Present CGPA: 3.32/4.00
Higher Secondary Certificate (HSC) Comilla Government College	GPA: 5.00
Secondary School Certificate (SSC) Comilla Modern High School	GPA: 5.00

KEY PROJECTS

MedRAG-VQA — Medical Image Visual Question Answering

Python · BiomedCLIP · LLaVA · LangChain · FAISS

- Built a multimodal RAG pipeline for clinical question answering on X-rays, MRIs, and CT scans using the SLAKE dataset
- Engineered fused image+text embeddings with BiomedCLIP and indexed them using FAISS for efficient retrieval
- Deployed LLaVA-1.5-7B (4-bit quantized) and integrated the full pipeline as a LangChain chain
- Significantly reduced hallucination with strong exact-match accuracy and BERTScore results

FAISS Similarity Search — Benchmarking & Analysis

Python · FAISS · NumPy · Matplotlib · Seaborn

- Benchmarked multiple FAISS indexes (FlatL2, IVFFlat, IVFPQ, HNSW, LSH) on SIFT1M dataset at 100K and 1M scales
- Analyzed build time, query latency, Recall@10, and memory usage with publication-quality visualizations
- Performed hyperparameter optimization and demonstrated clear speed-accuracy tradeoffs

IoT Smart Home Dashboard

Python · Tuya API · REST

- Developed real-time monitoring and control system for Tuya IoT devices

RibbonNet Detection — Self-Supervised Learning

BYOL · DINO · Swin-Tiny

- Implemented self-supervised representation learning using BYOL and DINO for robust medical image classification

Bone Fracture Detection System

Deep Learning · Computer Vision

- Developed CNN-based deep learning model for automated fracture detection in X-ray images

Book Recommender System

Collaborative Filtering · Content-based Filtering

- Built a hybrid recommendation engine using scikit-learn with comprehensive preprocessing and ranking metrics

Hospital Management System

Java · OOP · SOLID Principles

- Designed and implemented a full-featured hospital management system with patient registration, doctor dashboard, and appointment modules

Energy Monitoring Dashboard & Carbon Emission Calculator

Python · Streamlit · Render

- Built real-time energy tracking dashboard and web-based environmental impact calculator

ACTIVITIES AND COURSEWORK

Key Coursework

- Machine Learning, Deep Learning, Data Structures & Algorithms, Database Systems, Object-Oriented Programming, Software Engineering, Computer Networks

Core Competencies

- Multimodal AI & RAG Systems
- Computer Vision & Medical Imaging
- Vector Databases (FAISS)
- Deep Learning (CNNs, Vision Transformers)
- Recommendation Systems
- Data Analysis & Visualization
- Software Architecture & Design Patterns

- Prompt Engineering & LLM Integration

- System Design & UML Modeling

Leadership & Extracurricular

- Associate Executive, East West University Robotics Club — Conducted technical training for 200+ members and coordinated National Robo Fest 2024
- Leadership roles in Cumilla Govt. College Science Club (Event & Logistics Manager, Volunteer Commanding Officer)

Self-Directed Learning & Achievements

- Explored large language model fine-tuning and vector database technologies

RESEARCH PUBLICATION

IEEE Conference Paper (*Published — QPAIN 2025*)

<https://doi.org/10.1109/QPAIN66474.2025.11171763>

REFERENCES

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